

# **Requirements**

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In order to establish a strategy for planning and managing the project, the product brief served as the foundation for the team's requirements analysis.

Certain project requirements described in the brief were specific and allowed for an initial plan to be developed: conceptualising a 2d maze game to be built in a java based engine, with a 5 minute timer win condition and feature three different types of events. However it became clear that aspects of the brief were ambiguous, such as the physical attributes of the maze, events that hinder/boost the player, the theme and overall art style, game difficulty, the target market and accessibility expectations etc. This warranted further communication with the client regarding the brief and any specific user and system design preferences they may have.

Thus the team organised a series of questions to be delivered during an interview with the client, following the process of elicitation, analysis, and negotiation described by *Kolovos (2025)*.

These questions were divided into the following categories: resource limitations, project schedule and timeframe, risk management, the target audience and participants, modes of operation and constraints requirements. Examples of these updated requirements and their purposes will be presented throughout this document and supporting documentation found on the project website.

During interview preparation the team discovered the need for additional questions to clarify specific design and creativity limitations and requirement constraints. Particular focus was placed on difficulty implementation and complexity, as well as accessibility such as language support, target audience and competitiveness, since these were either loosely defined or missing from the product brief.

Clarification regarding the use of 3rd party assets and artificial intelligence generation/tools was also discussed, concluding that any AI generated content would only be acceptable in non-code or documentation areas such as art, audio, or textures, provided strict legal adherence to intellectual property and correct referencing was met. These requirements were then presented in a hierarchical structure of user, functional, and non-functional system requirements in the form of a referencing system, according to the **IEEE 29148-2018** guidelines to ensure clarity, traceability, and testability. This then allowed the team to allocate tasks to address each requirement and establish a more concrete plan for the project moving forward into the design phases.

## User Requirements

| ID              | Description                                                                                                                                    | Priority |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| UR_AUDIENCE     | The game should be targeted primarily towards students and adults, and be accessible to all.                                                   | Shall    |
| UR_GENRE        | The game should be a 2D maze-like puzzle game, with a focus on escape.                                                                         | Shall    |
| UR_EVENTS       | There should be a variety of interactable events that affect how the game plays.                                                               | Shall    |
| UR_DIFFICULTY   | The game should be aimed at a casual audience, and new players should be able to finish the game on their first playthrough.                   | Should   |
| UR_FEEDBACK     | The player will get feedback on how they performed after each playthrough.                                                                     | Shall    |
| UR_MARKETING    | The game will be family-friendly, containing no explicit or violent themes. No real-life people or things should be slandered within the game. | Shall    |
| UR-LEGAL        | All assets in the game must be legally sourced and appropriately licensed.                                                                     | Shall    |
| UR_OFFLINE      | The game should be a standalone application, and not rely on any online systems or resources.                                                  | May      |
| UR_ACHIEVEMENTS | There should be achievements that the player can complete as they play through the game.                                                       | Should   |
| UR_PROGRESSION  | The game should have a clear progression path for the player to follow, with a known end goal.                                                 | Should   |
| UR_IMMERSION    | The game shall offer a consistent and immersive user experience.                                                                               | Shall    |

## Functional System Requirements

| ID                 | Description                                                                                                                                             | User Requirements |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| FR_ACCESSIBILITY   | All visual cues and text boxes must be simple and easy to understand.                                                                                   | UR_AUDIENCE       |
| FR_MAP             | The game needs a 2D map in which the user can move, providing a boundary for the maze.                                                                  | UR_GENRE          |
| FR_EVENT_TYPES     | There should be multiple types of event that the player can interact with, all with different effects on the gamestate.                                 | UR_EVENTS         |
| FR_POSITIVE_EVENTS | Positive events should provide a fun boon of some kind to the player without making the experience too easy or boring.                                  | UR_EVENTS         |
| FR_NEGATIVE_EVENTS | Negative events must be manageable for the player and avoid excessive punishment or confusion.                                                          | UR_EVENTS         |
| FR_HIDDEN_EVENTS   | Hidden events should be invisible until triggered, at which point the player will know they have occurred.                                              | UR_EVENTS         |
| FR_PAUSE_MENU      | There must be an option for the user to pause the game, loading a pause menu. The game should transition seamlessly between paused and unpaused states. | UR_DIFFICULTY     |
| FR_WINNABLE        | There must always be at least one way to escape the maze, the player cannot be stuck in an unwinnable situation.                                        | UR_DIFFICULTY     |
| FR_END_SCREEN      | Upon finishing the game, an end screen should be displayed, with a time taken and a score shown.                                                        | UR_FEEDBACK       |
| FR_TIMER           | A countdown should be displayed while the game is running, telling the user how long the game has left.                                                 | UR_FEEDBACK       |

|                        |                                                                                                                                                                   |                 |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
|                        | The time should pause while the game pauses, and the user should learn the time taken to complete the game upon finishing.                                        |                 |
| FR_TIME_LIMIT          | The game should last five minutes                                                                                                                                 | UR_FEEDBACK     |
| FR_SCORE               | The game should display a score to the user upon finishing the game, which reflects how fast they were, with any adjustments caused by events taken into account. | UR_FEEDBACK     |
| FR_LEADERBOARD         | The game should display a ranked leaderboard of the top five scores when the game finishes.                                                                       | UR_FEEDBACK     |
| FR_ACHIEVEMENT_LIST    | The game should have a list of achievements that can be obtained by the player.                                                                                   | UR_ACHIEVEMENTS |
| FR_ACHIEVEMENT_TRIGGER | All achievements must have a trigger message or prompt informing the user that they have been completed.                                                          | UR_ACHIEVEMENTS |
| FR_GOAL                | The primary goal of the game is to escape the maze in the allotted time of 5 minutes. Maximising score should be a secondary goal for the user.                   | UR_PROGRESSION  |
| FR_SOUND               | There should be music and sounds that play throughout the game.                                                                                                   | UR_IMMERSION    |

## Non-Functional System Requirements

| ID                | Description                                                                                               | User Requirements |
|-------------------|-----------------------------------------------------------------------------------------------------------|-------------------|
| NFR_PERIPHERALS   | The game should not require the user to have any hardware beyond a keyboard and mouse.                    | UR_AUDIENCE       |
| NFR_ACCESSIBILITY | Accessibility features will be built in, and not require being toggled on.                                | UR_AUDIENCE       |
| NFR_CONTROLS      | The game should be operated using only a simple and intuitive set of controls.                            | UR_AUDIENCE       |
| NFR_LATENCY       | There should not be any delay between the user pressing an input and the game responding.                 | UR_AUDIENCE       |
| NFR_LANGUAGE      | The game and all text prompts for the user will be written in English.                                    | UR_AUDIENCE       |
| NFR_MAP_DESIGN    | The map should reflect a university campus, and should be in a 2D top-down view.                          | UR_GENRE          |
| NFR_INTRODUCTION  | The game should load into a brief introduction to the game and the game's controls.                       | UR_DIFFICULTY     |
| NFR_WIN_RATE      | The game should be completable after only one or two playthroughs.                                        | UR_DIFFICULTY     |
| NFR_LISCENSING    | All 3rd party assets and libraries must be fully compliant with intellectual property laws.               | UR-LEGAL          |
| NFR_AI_USAGE      | AI generated content can only be used in non-code or documentation areas such as art, audio, or textures. | UR-LEGAL          |
| NFR_APPLICATION   | The game should be executed as a standard PC application available on Windows.                            | UR_OFFLINE        |
| NFR_BOUNDARIES    | The boundaries of the maze should be clear and intuitive.                                                 | UR_PROGRESSION    |

|                 |                                                      |              |
|-----------------|------------------------------------------------------|--------------|
| NFR_PERFORMANCE | The game should run consistently at around 30-60fps. | UR_IMMERSION |
|-----------------|------------------------------------------------------|--------------|

## References

IEEE (2018) *ISO/IEC/IEEE 29148:2018 – Systems and software engineering – Life cycle processes – Requirements engineering*. IEEE Standards Association. Available at: <https://standards.ieee.org/standard/29148-2018.html> (Accessed: October 2025).

Kolovos, D. (2025) *Requirements Engineering* [video recording]. Engineering 1 module, University of York Virtual Learning Environment (VLE), accessed October 2025.